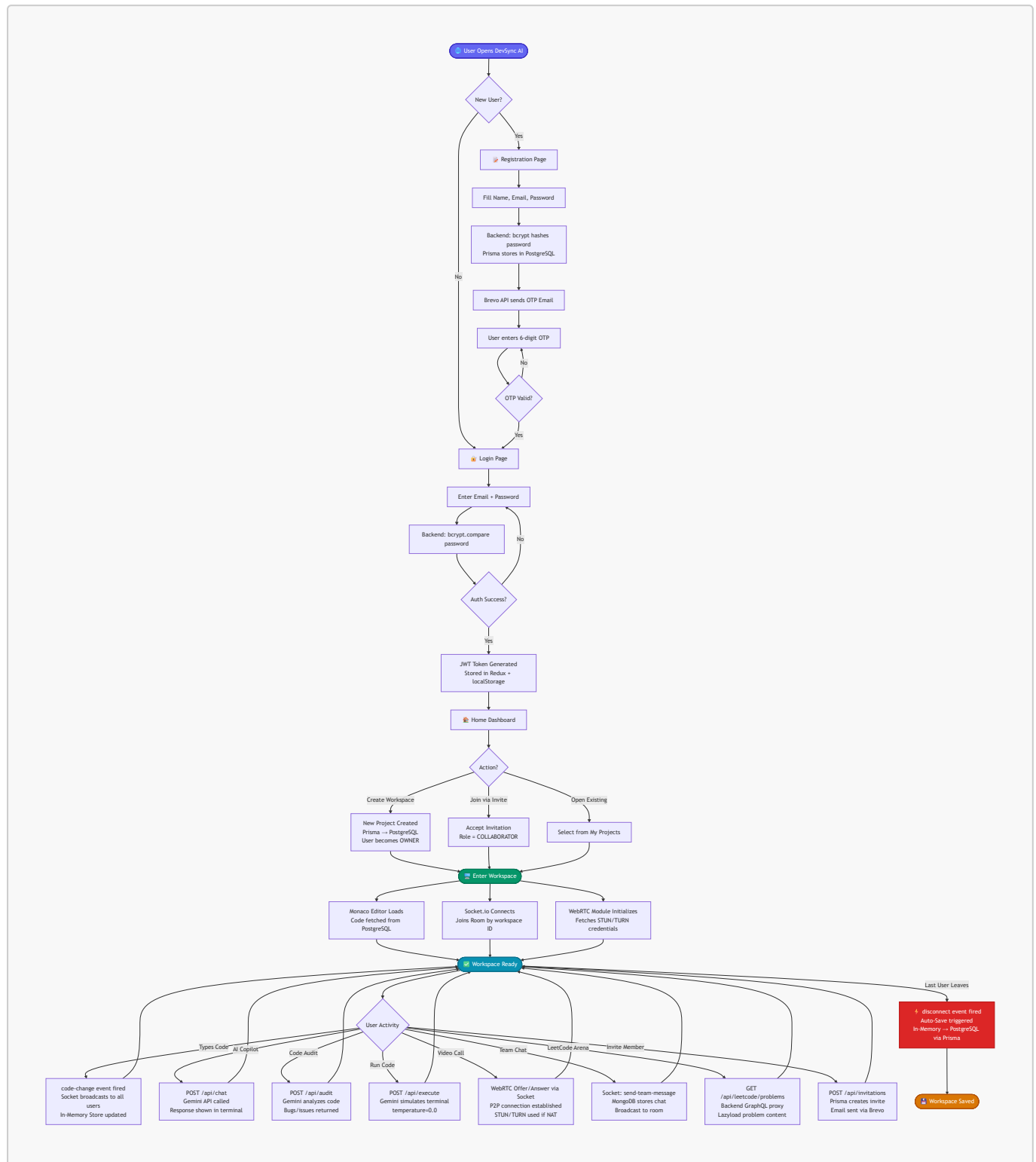


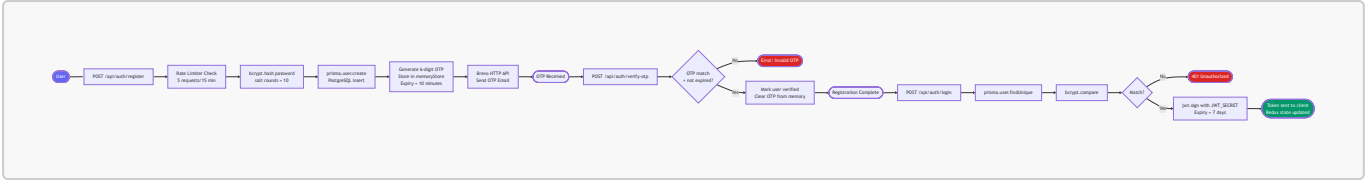
DevSync AI — Complete Project Workflow Flowchart

For Interview Explanation (Hinglish Talking Points)

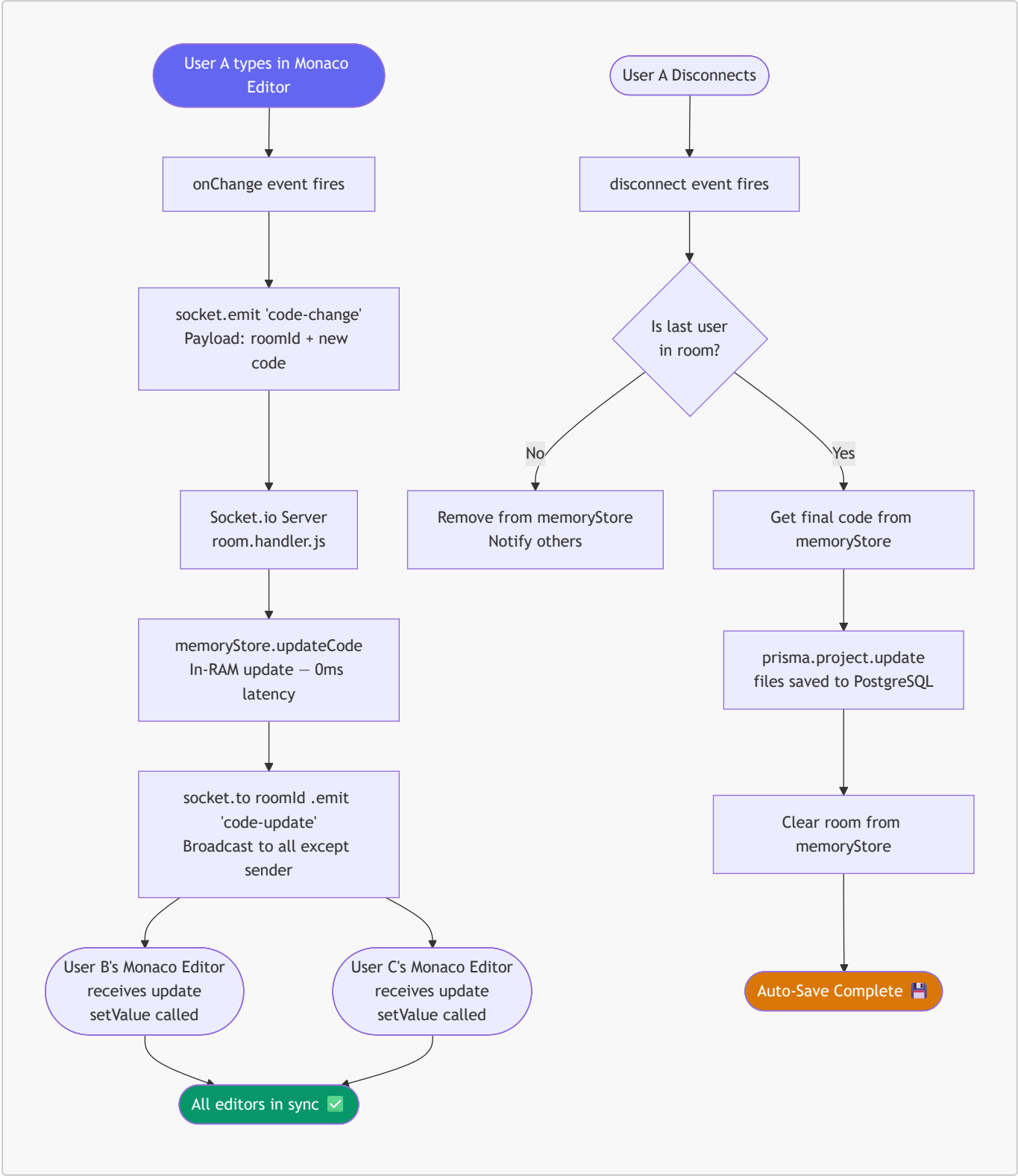
📖 MASTER FLOWCHART — Complete User Journey



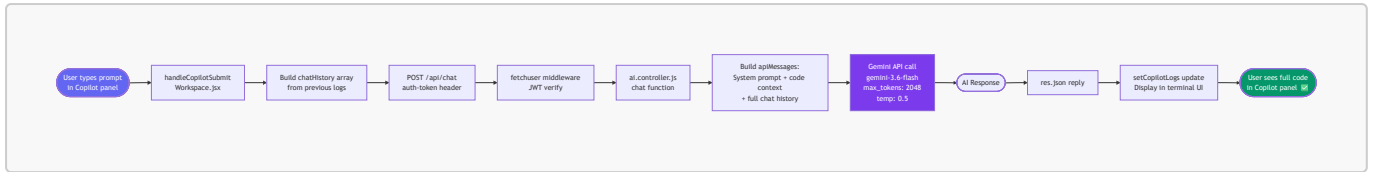
🔒 FLOWCHART 1 — Authentication Flow (Detail)



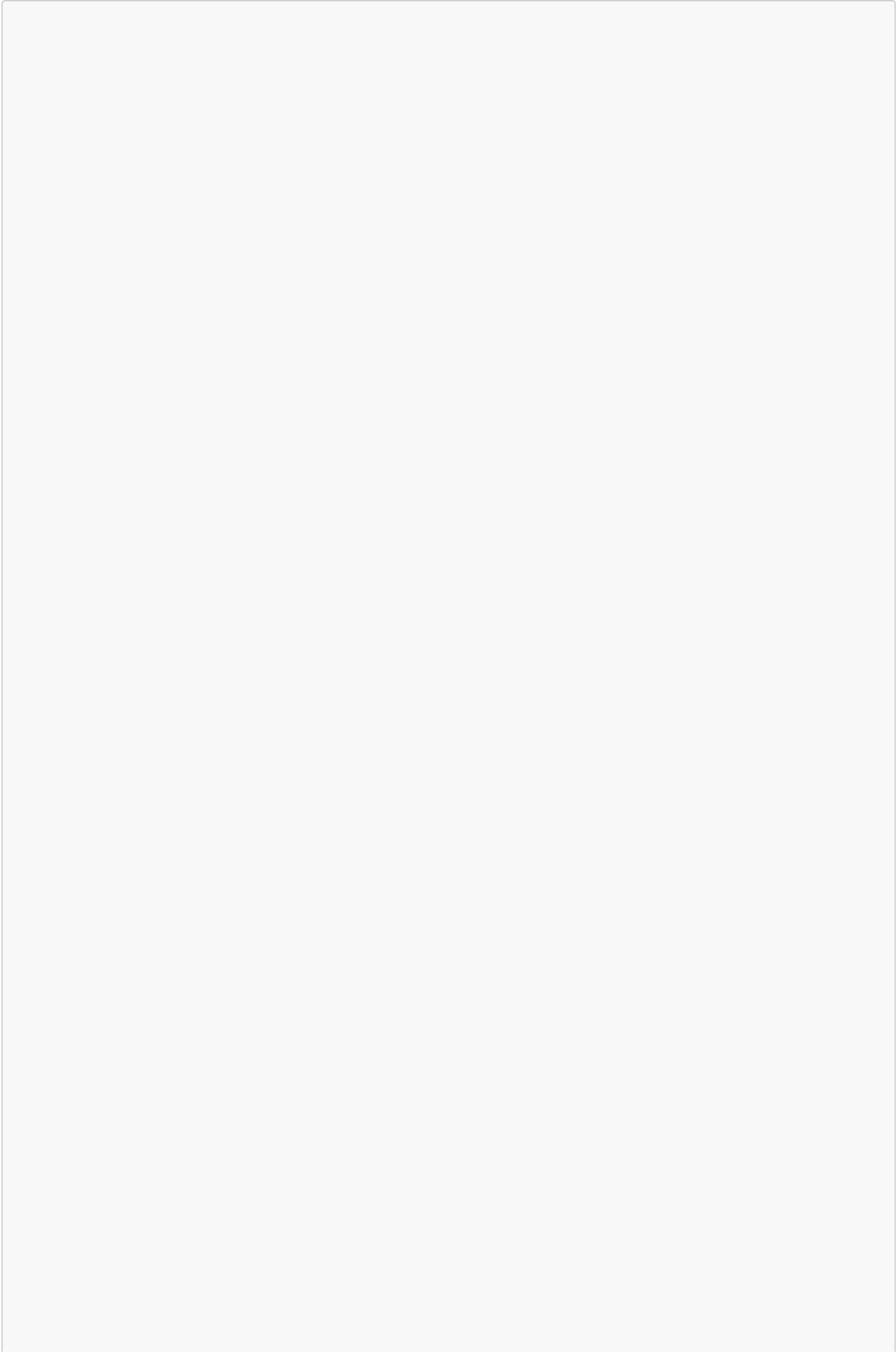
⚡ FLOWCHART 2 — Real-Time Code Sync Flow

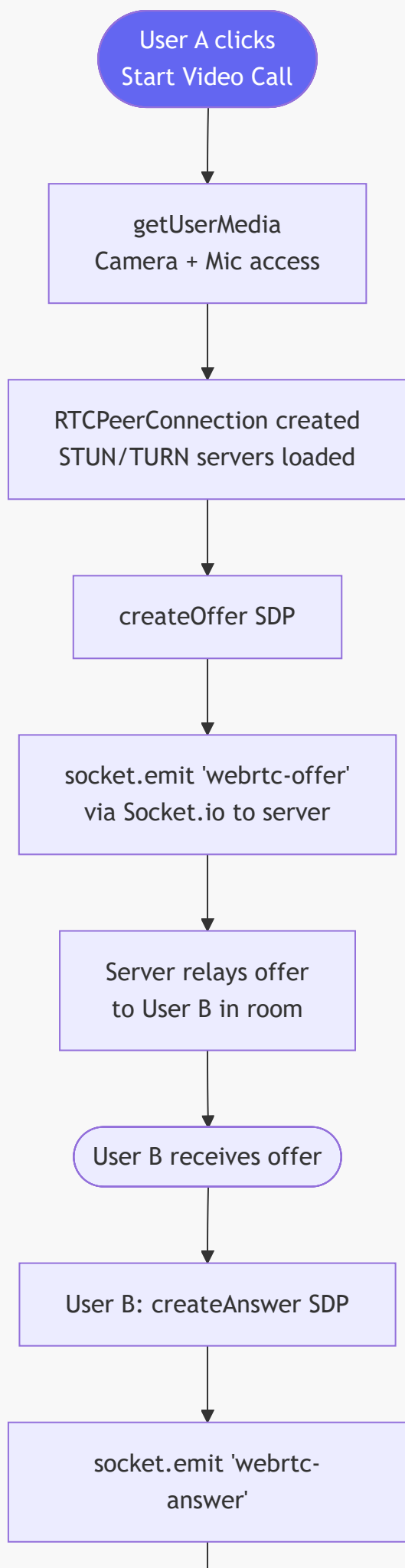


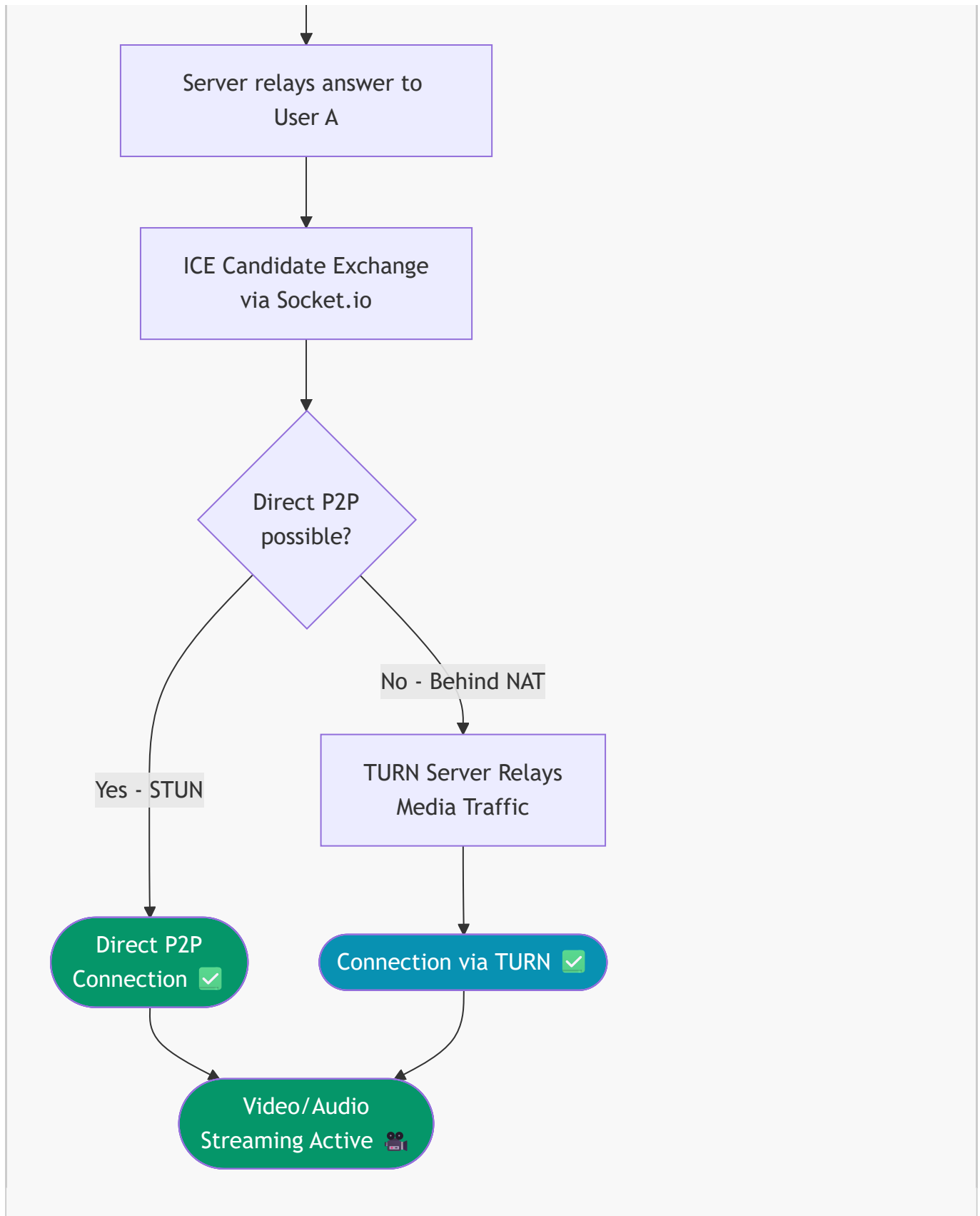
🤖 FLOWCHART 3 — AI Copilot Flow



FLOWCHART 4 — WebRTC Video Call Flow







INTERVIEW TALKING SCRIPT — Step by Step

Step 1: Jab Interviewer puche "Project explain karo"

"Sir, DevSync AI ek real-time collaborative IDE hai. Main aapko iske complete workflow ke through walk karta hoon."

📌 PHASE 1 — User Registration & Auth

Kya bolna hai:

- **"Pehla step user registration hai."**
- "Jab user sign up karta hai, uska password **bcrypt** se **hash** hota hai — plain text kabhi store nahi hoti."
- "Phir **Prisma ORM** se PostgreSQL mein **user.create** call hoti hai."
- "Ek 6-digit **OTP** generate hota hai jo **memoryStore** (server ki RAM) mein temporarily 10 minutes ke liye store hota hai."
- "Ye OTP **Brevo HTTP API** se email pe bheja jata hai — SMTP nahi use kiya kyunki Render jaise cloud platforms SMTP ports block karte hain."
- "OTP verify hone ke baad JWT token generate hota hai jo client side pe **Redux state** aur **localStorage** mein store hota hai."

🔑 **Key Terms to use:** **bcrypt**, **Prisma ORM**, **memoryStore**, **Brevo HTTP API**, **JWT**, **Redux**

📌 PHASE 2 — Workspace Entry

Kya bolna hai:

- **"Jab user workspace open karta hai, teen cheezein simultaneously initialize hoti hain."**
- **"Pehla:** Monaco Editor load hota hai — code PostgreSQL se fetch karke editor mein set hota hai."
- **"Doosra:** Socket.io connection establish hoti hai. Client **socket.emit('join-room')** karta hai apne workspace ID ke saath."
- **"Teesra:** WebRTC module initialize hota hai — STUN/TURN server credentials backend se fetch hote hain."

🔑 **Key Terms to use:** **Monaco Editor**, **Socket.io**, **join-room event**, **WebRTC**, **STUN/TURN**

📌 PHASE 3 — Real-Time Code Collaboration

Kya bolna hai:

- **"Ye project ka core feature hai."**
- "Jab User A type karta hai, Monaco Editor ka **onChange** event fire hota hai."
- "**socket.emit('code-change')** se server pe payload jata hai — **roomId** aur naya code."
- "Server side pe **room.handler.js** is event ko handle karta hai."
- "Code **In-Memory Store** (server ki RAM) mein save hota hai — iska fayda ye hai ki **database hit zero hoti hai** har keystroke pe."
- "Phir **socket.to(roomId).emit('code-update')** se baaki sab users ko update broadcast hota hai — **sender ko chhod ke.**"
- "Jab **last user disconnect** karta hai, Auto-Save trigger hota hai — RAM ka code **Prisma.update** se PostgreSQL mein permanently save ho jata hai."

🔑 **Key Terms to use:** **onChange event**, **code-change socket event**, **In-Memory Store**, **broadcast**, **Auto-Save on disconnect**, **Prisma.update**

📌 PHASE 4 — AI Copilot

Kya bolna hai:

- "AI Copilot context-aware hai — usse current file ka pura code milta hai."
- "Frontend se `POST /api/chat` call hoti hai — saath mein `auth-token` header mein JWT hota hai."
- "`fetchuser` middleware JWT verify karta hai."
- "`ai.controller.js` mein `System Prompt` build hoti hai — jisme active file ka code context inject hota hai."
- "Full conversation history bhi bheja jata hai taaki AI **thread context** maintain kar sake."
- "**Gemini 3.6 Flash** model call hota hai — `max_tokens: 2048` taaki poora code kabhi truncate na ho."
- "`temperature: 0.5` for chat — thoda natural lagay, `temperature: 0.0` for execution — deterministic output chahiye."

🔑 **Key Terms to use:** `System Prompt`, `context injection`, `conversation history`, `Gemini 3.6 Flash`, `max_tokens: 2048`, `temperature: 0.0 vs 0.5`

📌 PHASE 5 — WebRTC Video Calling

Kya bolna hai:

- "Video calling purely Peer-to-Peer hai — server sirf signaling karta hai."
- "User A `createOffer()` karta hai — ye **SDP (Session Description Protocol)** generate karta hai."
- "Offer Socket.io ke through server relay karta hai User B ko."
- "User B `createAnswer()` karta hai aur reply karta hai."
- "Dono ke beech **ICE Candidate exchange** hoti hai — ye process decide karta hai best connection route."
- "Agar direct connection possible ho toh **STUN** se public IP milti hai aur direct P2P hota hai."
- "Agar corporate firewall ya strict NAT ke piche ho, toh **TURN server** media relay karta hai."

🔑 **Key Terms to use:** `SDP`, `ICE Candidates`, `signaling via Socket.io`, `STUN`, `TURN`, `P2P`

📌 PHASE 6 — LeetCode Arena

Kya bolna hai:

- "LeetCode integration ke liye maine backend proxy banaya hai."
- "Directly frontend se LeetCode API call nahi ki — kyunki browser **CORS block** kar deta hai cross-origin requests."
- "Backend `leetcode.controller.js` mein GraphQL query run hoti hai LeetCode ke API pe."
- "Response filter karke clean data frontend bheja jata hai."
- "Problems **Lazy-Loaded** hain — initially sirf list aati hai, problem ka HTML content tabhi fetch hota hai jab user select kare."

🔑 **Key Terms to use:** `CORS bypass`, `Backend GraphQL Proxy`, `Lazy Loading`, `leetcode.controller.js`

📌 PHASE 7 — Security Architecture

Kya bolna hai:

- "Security ke liye maine 4 layers implement ki hain."
- "Layer 1: `bcrypt` — password hashing with salt."
- "Layer 2: `JWT` — stateless authentication, har protected route pe `fetchuser` middleware verify karta hai."
- "Layer 3: `RBAC` — Owner vs Collaborator roles, `checkMembership` middleware enforce karta hai."
- "Layer 4: `Rate Limiting` — `express-rate-limit` se DDoS aur brute-force attacks se protection."

🔑 **Key Terms to use:** `bcrypt` `salt`, `JWT` `stateless` `auth`, `fetchuser` `middleware`, `RBAC`, `checkMembership`, `express-rate-limit`

💡 CLOSING LINE FOR INTERVIEW

"To summarize — DevSync AI ek full-stack, production-ready platform hai jo real-time collaboration ke liye Socket.io, video communication ke liye WebRTC, data persistence ke liye dual-database architecture (PostgreSQL + MongoDB), aur AI features ke liye Google Gemini use karta hai. Sab kuch ek cohesive, secure, aur scalable architecture mein integrated hai."

Made for DevSync AI Interview Preparation